Assignment 1:
Design and Prototype a Learning CUI for the AI Era

Design and prototype a CUI for one specific university learning task

Item	Details
Weight	16 points
Main report	Maximum 2,000 words
Video	Maximum 10 minutes; maximum 200 MB
Deadline	22/09/2026 (Recess Week) 23:59
Submission	Canvas: PDF + ZIP + MP4

1. Assignment at a glance

In this individual assignment, you will investigate one specific university learning task in which students use generative AI but still experience a learning difficulty. You will identify one evidence-supported difficulty, define a focused learning goal, decide what the learner and the CUI should each do, choose a suitable CUI form and technical approach, and develop a prototype of one core, multi-turn CUI learning interaction. Assignment 1 prepares this interaction for implementation in Assignment 2. You will demonstrate and explain your prototype in an individual video.

You will be assessed on the quality and justification of your decisions, how clearly your prototype communicates the intended interaction, and whether you understand and can explain what you submitted. You may use AI tools throughout the assignment. However, AI may support rather than replace the responsibilities listed below.

How you may use AI

The following responsibilities remain yours when you use AI:

1. You must personally obtain consent and talk to or observe at least three real adult university students. AI-generated or simulated participants do not count as learner evidence.
2. You must make the final decisions about the learning problem, learning goal, CUI role, and technical approach.
3. You must check that your prototype accurately represents the intended interaction and understand any code, prompts, configurations, and claims you submit.
4. You must appear in the video and explain the project in your own words. An AI-generated avatar or voice cannot replace you.

What counts as a CUI?

A CUI is not limited to a text chatbot. In this assignment, a CUI is an interactive system in which the learner and the system take turns communicating, and the system's next response or action is shaped by what the learner says, does, or provides.

It may use text, voice, video, multimodal interaction, external tools, or agents. It may act as a tutor, coach, practice partner, teachable peer, reflection partner, or another learning-oriented role.

Note:

1. Voice, video, multimodal, or agent-based systems do not receive automatic bonus marks compared to a text-only chatbot.
2. Choose the form that **best fits the learners**, learning goal, and what you can build well.
3. Using a larger, newer, or paid model is not, by itself, an innovation.

Examples for inspiration:

- **StuBot** asks learners to teach a topic in their own words and responds with questions about missing parts.
- **Khanmigo** acts as a coach that guides learners instead of simply giving them an answer.
- **Duolingo Roleplay and Video Call** provide conversational language practice followed by feedback.

2. What you need to do

Complete the following four steps in order. Use the same order in your report.

Step 1: Understand what remains difficult after AI assistance

Interview at least **3 adult university students** about **one specific learning task** (e.g., programming, reading, mathematics, abstraction, planning, and decomposition) for which they currently use generative AI. You may ask them questions or observe how they use and respond to the AI's help. Ask for their permission and do not include names or other identifying information in your submission.

Summarise what you learned in four parts:

1. **Learner and task:** Who (major? Grad? … etc) are you designing for, what are they trying to learn, and how do they currently use AI?
2. **Current difficulty:** After receiving AI assistance, what does the learner still struggle to understand, or do for themselves?

   Choose one of the following difficulties that best describes your target learner, or identify another difficulty supported by your learner evidence:

   - **Reduced persistence:** When the task becomes difficult, the learner turns to AI before making or sustaining a meaningful attempt of their own.
   - **Uncritical use of AI output:** The learner can obtain an AI-assisted answer but struggles to explain, question, verify, or improve it using their own reasoning.
   - **Reduced ownership and self-direction:** The learner relies on AI to determine the approach or next steps and struggles to make key decisions for themselves.
   - **Other:** A different difficulty that emerged from your interviews or observations and is not adequately captured by the categories above. Give it a clear name and definition.

   For your chosen difficulty, describe how it appears in the specific learning task you are studying. Explain what the learner can accomplish with AI assistance and what they still struggle to understand or do on their own. Your choice must be supported by evidence from your learner interviews or observations.
3. **User Needs:** Summarize key needs identified from target user interviews
4. **Potential CUI support:** How a CUI could assist them in overcoming these obstacles, and how it can empower them to master these tasks independently.

Step 2: Decide the learning goal and the CUI's role

Describe your plan in the following parts:

1. **Learning goal:** Based on the difficulty identified in Step 1, state one specific thing the learner should understand or learn to do. Explain what the learner would explain or do to show progress toward this goal.
2. **Roles of the learner and the CUI:** Clearly state what the learner must think about or do themselves during the learning activity, and what the CUI will do to support that work without taking it over.
3. **Why this approach:** What does conversational interaction add to this learning process? Could a simpler non-conversational design work just as well?
4. **Design Goals:** Define at least three clear design goals grounded in relevant theories and identified user needs, providing a solid justification for each.

Step 3: Implementation Plan

Detail the CUI model and tech stack (such as a chatbot, hybrid CUI+GUI, multimodal framework, rule-driven logic, or prompt engineering) chosen for your functional prototype, explaining how these options directly support the target learning challenge, learning objective, and planned interaction model.

Outline your implementation strategy. While implementation is not required for Assignment 1, and prototyping tools are permitted, you must clearly demonstrate your technical plan and understanding:

- What components need to be developed?
- Which parts will you build yourself?
- Which aspects will rely on an LLM for development assistance?

Provide a clear overview of the system architecture and justify your approach to leveraging AI to build the conversational interface. Keep in mind that technical complexity alone does not translate to higher marks. A straightforward CUI can earn top marks if it addresses student challenges effectively and accurately. Ultimately, success hinges on sound design decisions rather than reliance on complex or cutting-edge technologies.

Step 4: Develop a CUI prototype

Develop a prototype of one multi-turn CUI learning interaction based on your work in Steps 1–3. The interaction must involve several connected exchanges between the learner and the CUI as they work toward the learning goal. Assignment 1 focuses on designing and prototyping this interaction to be ready for implementation, not on building a CUI system. You may use code, digital prototyping tools, storyboards, paper materials, or another suitable prototyping method. Whatever method you choose, the prototype must be detailed enough to demonstrate how the interaction works and how it addresses one key learning need identified in Step 1.

Your prototype does not need to cover every planned feature, but it must:

1. Enable the learner to engage in and complete a single core learning activity across multiple conversational turns.
2. Show what the learner thinks about or does, and how the CUI's responses are shaped by the learner's input across the interaction.
3. Show how the interaction addresses the identified learning need and supports the learning goal from Step 2.

In your demonstration video, explicitly show how the interaction works:

4. Walk through the interface to illustrate the system operations and expected user interactions (imperfections in an initial prototype are completely fine).
5. Roleplay or act out the step-by-step user interaction to show how the system is intended to function.

3. What to submit

A report PDF, a prototype package ZIP, and an individual video MP4. Use the filenames below and replace <StudentID> with your own student ID.

A. A1_<StudentID>_Report.pdf

Maximum 2,000 words; no minimum. Ensure the report remains concise by focusing strictly on content directly relevant to explaining your project. The 2,000-word limit covers the core text only, excluding references and optional appendices. Structure the report in the same order as Steps 1–4 above.

Cite all external research, code, datasets, media, templates, and other resources used.

B. A1_<StudentID>_Prototype.zip

Your prototype package must contain:

1. All materials needed to review and demonstrate your prototype, such as source code, prompts, configurations, digital prototype files, storyboards, or clear scans or photographs of paper materials.
2. A README.md that identifies the prototyping method and explains how to open, run, view, or roleplay the prototype.

C. A1_<StudentID>_Video.mp4

Submit an individual MP4 video of no more than 10 minutes and no more than 200 MB.

1. Show your face while speaking and keep it visible while you explain the project.
2. Demonstrate one complete, multi-turn CUI learning interaction from beginning to end. If the prototype is paper-based or simulated, role-play the interaction.
3. Using your own words, explain the learning problem, how the learner and the CUI divide the work, one important design decision, and one current limitation.

Evaluations focus on a straightforward demonstration, a clear rationale tied to your project, and evidence of your genuine comprehension and ownership of the work. Video editing is not necessary—a basic Zoom recording is fully sufficient. Submissions are not judged on visual polish; instead, the focus is on how effectively you articulate your concept and convey your underlying thought process throughout the prototyping phase.

4. How your work will be assessed

Your Assignment 1 mark is out of 16 and counts directly toward the 100-mark course total. Graders will assess the four areas below.

Learning Problem and User Needs (**4.0 marks**)
To demonstrate a deep understanding of the problem space and target user group, base your submission on primary empirical evidence collected through interviews or observational sessions with at least 3 adult university students. Identify their specific academic program, gender, or year of study rather than names or IDs to demonstrate relevance to your chosen topic. You must clearly outline a focused learning task and examine how students currently incorporate generative AI tools into their study workflow. Pinpoint a primary challenge

faced by learners, supporting your findings with direct user quotes, observed behaviors, or documented responses. Conclude by articulating the fundamental user need your CUI design intends to address.

Learning Design and Chosen Approach (**4.0 marks**)
This section requires a well-defined learning objective alongside a clear mechanism for tracking student progress. You should establish an explicit division of responsibilities between the student and the CUI, while providing a compelling rationale for adopting a conversational modality. Your design goals must be rooted in established communication and learning theories as well as empirical user needs. Furthermore, present a feasible, well-suited technical implementation strategy and CUI framework, ensuring every design choice and architectural decision is thoroughly justified. To ground your approach, draw upon lecture materials, theoretical foundations, and relevant scholarly literature from the course reading list or external academic sources.

Please properly attribute or cite these resources in your report, so then we will be able to know how you grounded your design properly.

CUI Prototype and Interaction (**3.0 marks**)
Construct a low-fidelity prototype illustrating a multi-turn CUI learning exchange. The design must foster active cognitive engagement from the learner, offer tailored feedback, guidance, or progressive steps following user input, and directly address the target learning challenge. Evaluation centers on the clarity and coherence with which the interaction design is communicated. Suitable medium formats encompass low-complexity solutions such as conversation flows, paper-based mockups, scripts, storyboards, or Dialogflow setups. Selection of the appropriate prototyping approach is left to your discretion.

The purpose of these rapid prototyping techniques is to clarify the core concept and prepare it for implementation and user testing in Assignment 2. High visual fidelity is not expected; assessment focuses on the quality and clarity of the interaction design rather than its visual presentation.

Individual Video and Ownership (**5.0 marks**)
A clear demonstration of the complete, multi-turn prototype interaction; project-specific explanation and reasoning; and clear evidence that the student understands and owns the submitted work. Within this criterion, 2 marks are for the demonstration, 2 marks are for project-specific explanation and reasoning, and 1 mark is for individual understanding and ownership.